

CS348 Project Stage 3 Report

Github Link: <https://github.com/Aaron0825/CS348-Project>

Indexes

I use unclustered B+ tree indexes for this project, not also because this is the default index for Django and Postgre but also because some of my queries are range queries. I have 10 indexes in total, which are all defined in models.py in the respective models (tables).

In Student, I have an index for students' first name and last name, which is useful for the search box in the Classlist page for teachers.

In Test, I have an index for test date, which is useful for the date range search in the Tests page for teachers.

In TestResult, I have an index for student, an index for test, and a composite index for (student, test). These indexes are useful in two places: the detail buttons on Classlist page and Test page to query students who took a specific test or all the tests a specific student took.

In Homework and Homework Result are the rest of indexes, which works exactly the same as those in Test and TestResult, but this time on Classlist and Assignment pages.

Concurrency

The isolation level I choose for this application is READ COMMITTED, for this app is designed for one course (cs 348) only, there are only around 250 students and a handful of instructors.

Students only do SELECTs on their own grades (read only, no write) – they don't need to see uncommitted data, and they don't need the overhead of stronger isolation. Instructors

occasionally INSERT/UPDATE/DELETE, and READ COMMITTED ensures they never see dirty rows from other transactions, yet avoids the locking and serialization overhead of higher levels; for there are no mass modify/delete function in this app, instructors don't need to worry about phantom data. Also I use transactions on all functions that involve add/update/delete data and not on those read-only functions, to ensure any modification on the database is always committed or aborted.